

Deciphering the genetic interactions between Pou4f3, Gfi1 and Rbm24 in maintaining cochlear hair cell survival

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Abstract

Mammals have limited sound receptor hair cells (HCs) that cannot be regenerated after damage. Thus, investigating the molecular mechanisms underlying how to maintain HC survival is crucial to prevent hearing impairment. The *Pou4f3*^{-/-} or *Gfi1*^{-/-} HCs initially form but rapidly degenerate, whereas *Rbm24*^{-/-} HCs degenerate much later. It remains elusive about the transcriptional cascades among Pou4f3, Gfi1 and Rbm24. Here, we demonstrate that *Rbm24* expression is completely repressed in *Pou4f3*^{-/-} HCs, but its expression is not altered in *Gfi1*^{-/-} HCs. Moreover, both Pou4f3 and Gfi1 expressions are intact in *Rbm24*^{-/-} HCs. Moreover, by the *in vivo* mouse transgenic reporter assays, we identify three *Rbm24* enhancers to which Pou4f3 binds. Finally, we test whether Rbm24 restoration can alleviate degeneration of *Pou4f3*^{-/-} HCs. Our *in vivo* genetic assay shows that ectopic Rbm24 alone is unable to stop *Pou4f3*^{-/-} HCs from undergoing degeneration. Collectively, our study provides new molecular and genetic insights into how HC survival is regulated.

eLife assessment

In this **useful** study, the authors explore regulatory cascades governing mammalian cochlear hair cell development and survival. They confirm previous studies that the transcription factors Pou4f3 and Gfi1 are necessary for hair cell survival, and use **compelling** evidence to demonstrate that the RNA binding protein gene RBM24 is regulated by Pou4f3, but not Gfi1. These findings will be of interest to those working on hearing loss, and hold significance for viral gene delivery methods aiming to manipulate gene expression.

Introduction

Housed in the mammalian cochlea, the auditory epithelium, also referred to as the Organ of Corti (OC), contains the mechanosensory receptors hair cells (HCs) that detect sound information ^{1,2}. Two types of HCs exist: inner HCs (IHCs) and outer HCs (OHCs), with one row of IHCs and

three rows of OHCs ³, OHCs and IHCs derive from the cochlear progenitors experiencing high level of *Atoh1*, a master transcription factor (TF) in HC development ^{4–6}. No HCs are formed in *Atoh1*^{−/−} cochleae, highlighting the essential roles of *Atoh1* in specifying the general HC fates in the undifferentiated progenitors ^{7, 8}. OHCs uniquely express motor protein Prestin (encoded by *Slc26a5*) and serve as sound amplifiers ^{9, 10}. In contrast, IHCs specifically express *Fgf8*, *vGlut3* (encoded by *Slc17a8*) and *Otoferlin* ^{11–15}, acting as the primary sensory cells that form synapses with different subtypes of type I spiral (auditory) neurons ^{16–19}. *Insm1* is necessary in stabilizing OHC fate, and ~50% of *Insm1*^{−/−} OHCs tend to transdifferentiate into IHC-like cells ^{20, 21}. *Ikzf2* is not needed in early OHC development but is required in consolidating the OHC fate, and *Ikzf2*^{cello/cello} deficient OHCs are dysfunctional and misexpress IHC genes ²². *Insm1* is an epistatic but indirect positive regulator of *Ikzf2*, and restoration of *Ikzf2* can partially mitigate the phenotypes of *Insm1*^{−/−} OHCs ²¹. In contrast to *Insm1* and *Ikzf2*, *Tbx2* is required in IHC fate specification, differentiation, and maintenance, and IHC to OHC conversion occurs in the absence of *Tbx2* ^{23–25}.

It is necessary to maintain the survival of cochlear HCs after their production. *Pou4f3*, a POU-domain family TF and known hearing loss gene (DFNA15), is essential for HCs to survive ^{26–30}. Different from *Atoh1*, *Pou4f3* is not involved in fate determination of HCs ³¹. As one target gene regulated by *Atoh1* ^{27, 32}, *Pou4f3* is turned on in HCs between E14.5 and E16 in a basal to apical gradient, and subsequently *Pou4f3* activates *Gfi1*, another zinc finger TF ^{33, 34}. *Gfi1* becomes undetectable in *Pou4f3*-deficient HCs ³⁴. HCs, especially the OHCs, degenerate in *Pou4f3* and *Gfi1*-mutants by birth ³⁴. Besides *Pou4f3* and *Gfi1*, RNA binding protein *Rbm24* is also specifically expressed in cochlear HCs ³⁵. Our recent study shows that *Rbm24* is indispensable in maintaining the survival of OHCs, but the OHC death happens much later than the case in *Pou4f3* and *Gfi1*-mutants, with OHCs develop normally by birth but the majority of them undergo cell death by postnatal day 19 (P19) in *Rbm24*^{−/−} mice ³⁶. It remains elusive about how *Rbm24* is regulated during cochlear HC development.

The similar but delayed phenotype of OHC death in *Rbm24* mutants relative to *Pou4f3* and *Gfi1*-deficient mice prompted us to dissect the potential genetic interactions among them by utilizing *in vivo* genetic approach. Our data clearly showed that onset of *Rbm24* expression was completely repressed in *Pou4f3*^{−/−} HCs. Surprisingly, *Rbm24* expression was normal in *Gfi1*^{−/−} HCs. Moreover, neither *Pou4f3* nor *Gfi1* expression was altered in the *Rbm24*^{−/−} HCs. Thus, our data showed that *Pou4f3*, but not *Gfi1*, is required in *Rbm24* expression. Furthermore, we identified three *Rbm24* enhancers that were sufficient to drive specific EGFP reporter expression in HCs, which are likely bound by *Pou4f3*. Finally, restoration of *Rbm24* alone cannot alleviate the degeneration of *Pou4f3*^{−/−} HCs, highlighting that additional targeted genes of *Pou4f3* are needed to be restored before keeping *Pou4f3*^{−/−} HCs able to survive. In sum, our study provides new insights into the genetic interactions among *Pou4f3*, *Gfi1* and *Rbm24*, with potential applications into HC protection.

Results

Rbm24 expression is completely repressed in *Pou4f3*^{−/−} cochlear HCs

HC degeneration happens in both *Pou4f3*^{−/−} and *Rbm24*^{−/−} mice, with the phenotypes occur earlier and are more severe in *Pou4f3*^{−/−} mice ^{34, 36}. It prompted us to speculate that genetic interaction between *Pou4f3* and *Rbm24* exists. Given *Pou4f3* is the upstream positive regulator of *Rbm24*, *Rbm24* expression should be blocked in the absence of *Pou4f3*. To rapidly determine whether it is the case, we took advantage of our previously established CRISPR-stop approach ³⁷. As illustrated in **Supplemental Figure 1A**, co-injecting one sgRNA-1 against *Pou4f3* and base editor components in one-cell stage mouse zygotes yielded founder 0 (F0) mice whose tail DNA

was subjected to sanger sequencing. Relative to wild type (WT) mice (**Supplemental Figure 1B**), the F0 mice carrying the homozygous premature emergence of stop codon (TAG) were defined as *Pou4f3*^{-/-} mice that were immediately ready for phenotypic analysis (**Supplemental Figure 1C**).

Triple staining of *Pou4f3*, *Rbm24* and *Insm1* demonstrated that OHCs were *Pou4f3*⁺/*Rbm24*⁺/*Insm1*⁺, and IHCs expressed *Pou4f3* and *Rbm24*, but not *Insm1* in basal turn of WT mice at E16.5 (n=3, **Supplemental Figure 1D-D**). *Pou4f3* expression was completely absent in the *Pou4f3*^{-/-} mice (n=9, **Supplemental Figure 1E-E**), confirming that *Pou4f3* translation was blocked. Different from WT mice, *Rbm24* expression completely disappeared in the *Pou4f3*^{-/-} mice at E16.5 (**Supplemental Figure 1E-E**). Notably, *Insm1* expression seemed normal in *Pou4f3*^{-/-} mice (arrows in **Supplemental Figure 1E-E**), although we noticed that the alignment of *Insm1*⁺ OHCs was not as regular as the WT OHCs (**Supplemental Figure 1D-D**). Moreover, the presence of *Insm1*⁺ OHCs in the *Pou4f3*^{-/-} mice ruled out the possibility that the absence of *Rbm24* was due to a secondary effect of OHC death or delayed differentiation, and also agreed with previous report that initial HC differentiation can occur without *Pou4f3*^{31, 34}. Thus, the rapid *Pou4f3* loss-of-function analysis by CRISPR-stop supports our hypothesis that *Pou4f3* is required in turning on *Rbm24* expression.

To further validate the observation above, we also constructed a germ line stable *Pou4f3* mutants in which the entire *Pou4f3* genome between sgRNA-2 and sgRNA-3 was deleted (**Figure 1A**). The three genotypes of *Pou4f3*^{+/+}, *Pou4f3*^{+/-} and *Pou4f3*^{-/-} were readily distinguished by tail DNA PCR (**Figure 1B**). Relative to WT mice (n=3, **Figure 1C**), *Rbm24* again was completely absent in *Pou4f3*^{-/-} mice at E16.5 (n=3, **Figure 1D**). Like *Insm1*, *Bcl11b* is another OHC marker²⁰. Triple staining of *Rbm24*, *Pou4f3* and *Bcl11b* also demonstrated that, relative to WT HCs (**Figure 1E-E**), *Rbm24* was completely absent in *Pou4f3*^{-/-} HCs (**Figure 1F-F**). Presence of *Bcl11b*⁺ OHCs also confirmed that production of nascent OHCs were normal in the absence of *Pou4f3*. Moreover, in contrast to WT mice at P1 (n=3, **Supplemental Figure 2A-A**), HC degeneration was manifest in the germ line *Pou4f3*^{-/-} mutants at P0 (n=4, **Supplemental Figure 2B-B**) and P1 (n=3, **Supplemental Figure 2C-C**). By normalizing numbers of the remaining HCs in *Pou4f3*^{-/-} mice at P0 or P1 to those in WT mice at P1, we found that in P0 *Pou4f3*^{-/-} mice, 35.94% ± 4.52%, 38.16% ± 2.79% and 63.8% ± 5.28% HCs survived in basal, middle and apical turns, respectively. However, in P1 *Pou4f3*^{-/-} mice, 5.54% ± 3.11%, 15.79% ± 3.65% and 65.64% ± 7.95% of the HCs respectively survived in basal, middle and apical turns. Thus, HC degeneration at middle and basal turns was more severe (** p<0.01) at P1 than that at P0, but there was no significant difference in apical turns (**Supplemental Figure 2D**).

Different from WT cochleae in which all *Myo7a*⁺ HCs expressed *Rbm24* (**Supplemental Figure 2E-E**), in *Pou4f3*^{-/-} cochleae the remaining HCs lost *Rbm24* expression (arrows in **Supplemental Figure 2F-F**). Collectively, our data from two different *Pou4f3*-deficient models support that *Pou4f3* is indispensable for turning on *Rbm24*. Nonetheless, whether *Gfi1* is also essential for *Rbm24* expression remains to be determined.

Construction of a *Gfi1-3× HA-P2A-Cre*^{+/+} knockin mouse strain

We determined whether the regulation of *Rbm24* by *Pou4f3* through *Gfi1* or independently of *Gfi1*. Due to lack of a good commercial *Gfi1* antibody suitable for immunostaining, we decided to construct a *Gfi1-3× HA-P2A-Cre*^{+/+} (*Gfi1*^{HA-Cre}^{+/+} in brief) knockin mouse strain via our routine CRISPR/Cas9 approach (**Supplemental Figure 3A-C**). In principle, the C-terminus of *Gfi1* was tagged with three HA fragments, and the *Cre* is under the endogenous *cis*-regulatory elements of *Gfi1*. The WT (*Gfi1*^{+/+}), heterozygous (*Gfi1*^{HA-Cre}^{+/+}) and homozygous (*Gfi1*^{HA-Cre}^{HA-Cre}) mice were readily distinguished by tail PCR (**Supplemental Figure 3D**). In addition to the *Gfi1* locus, Southern blot assay showed that random insertion of the targeting vector (**Supplemental Figure 3B**) happened in the unknown genomic region. Nonetheless, it was likely that the random insertion happened in a silent genomic region, which was evidenced by the analysis below.

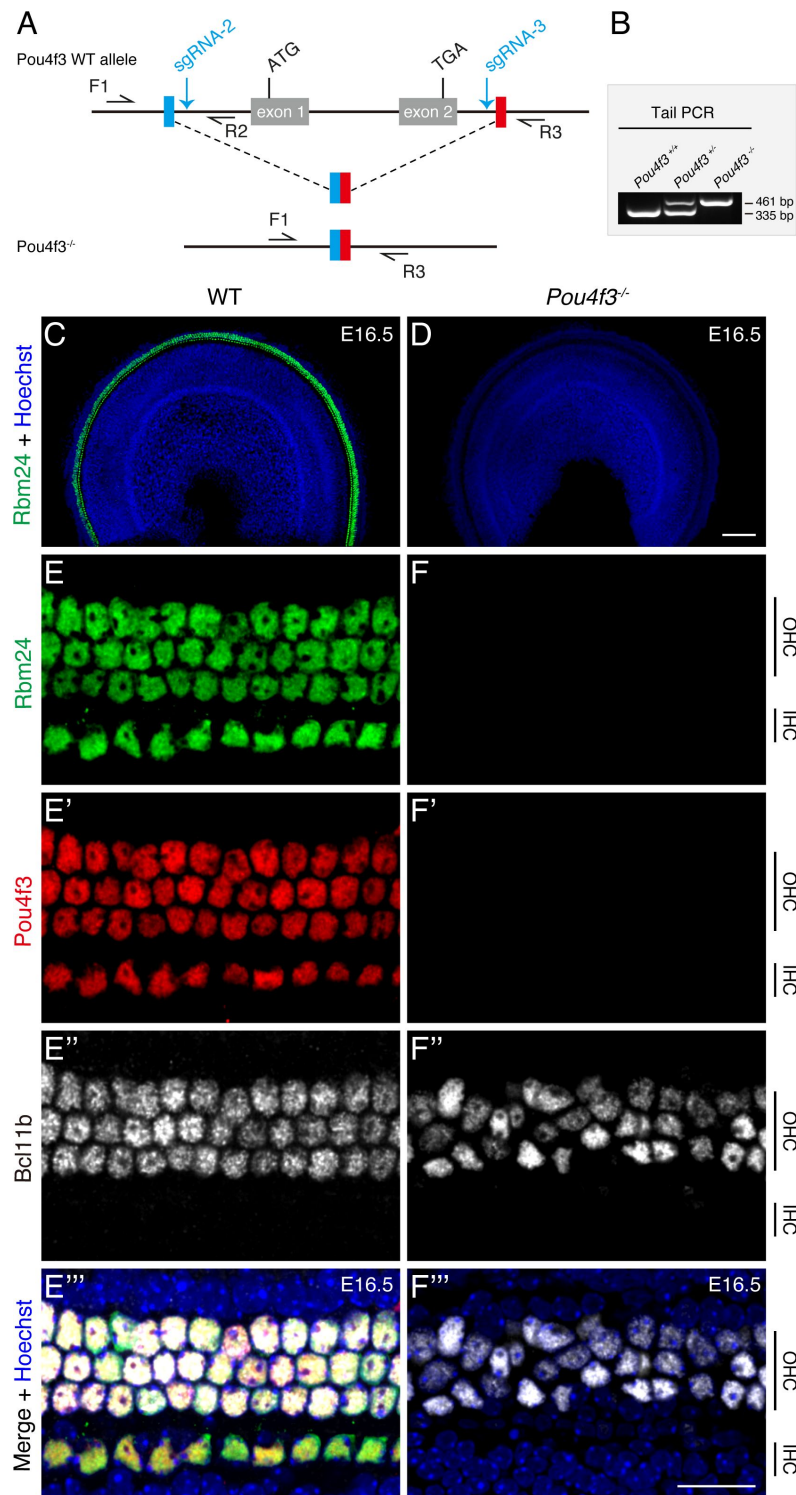


Figure 1.

Rbm24 expression is repressed in cochlear HCs in the absence of Pou4f3.

(A) A simple illustration of how *Pou4f3*^{-/-} strain was generated. **(B)** One exampled tail PCR gel image that helped us to distinguish WT (*Pou4f3*^{+/+}), heterozygous (*Pou4f3*^{+/-}) and homozygous (*Pou4f3*^{-/-}) mice. **(C-F'')** Triple staining of Rbm24, Pou4f3 and Bcl11b in WT (C and E-E'') and *Pou4f3*^{-/-} (D and F-F'') mice (n=3 for each) at E16.5, with only Rbm24 (green) channel was visualized in (C and D) at low magnification. (E-E'') and (F-F'') were the high magnification images of middle turn WT and *Pou4f3*^{-/-} mice, respectively. In the absence of Pou4f3, Rbm24 expression completely disappeared, despite nascent Bcl11b+ OHCs were present. OHC: outer hair cell; IHC: inner hair cell. Scale bars: 100 μm (D), 20 μm (F'').

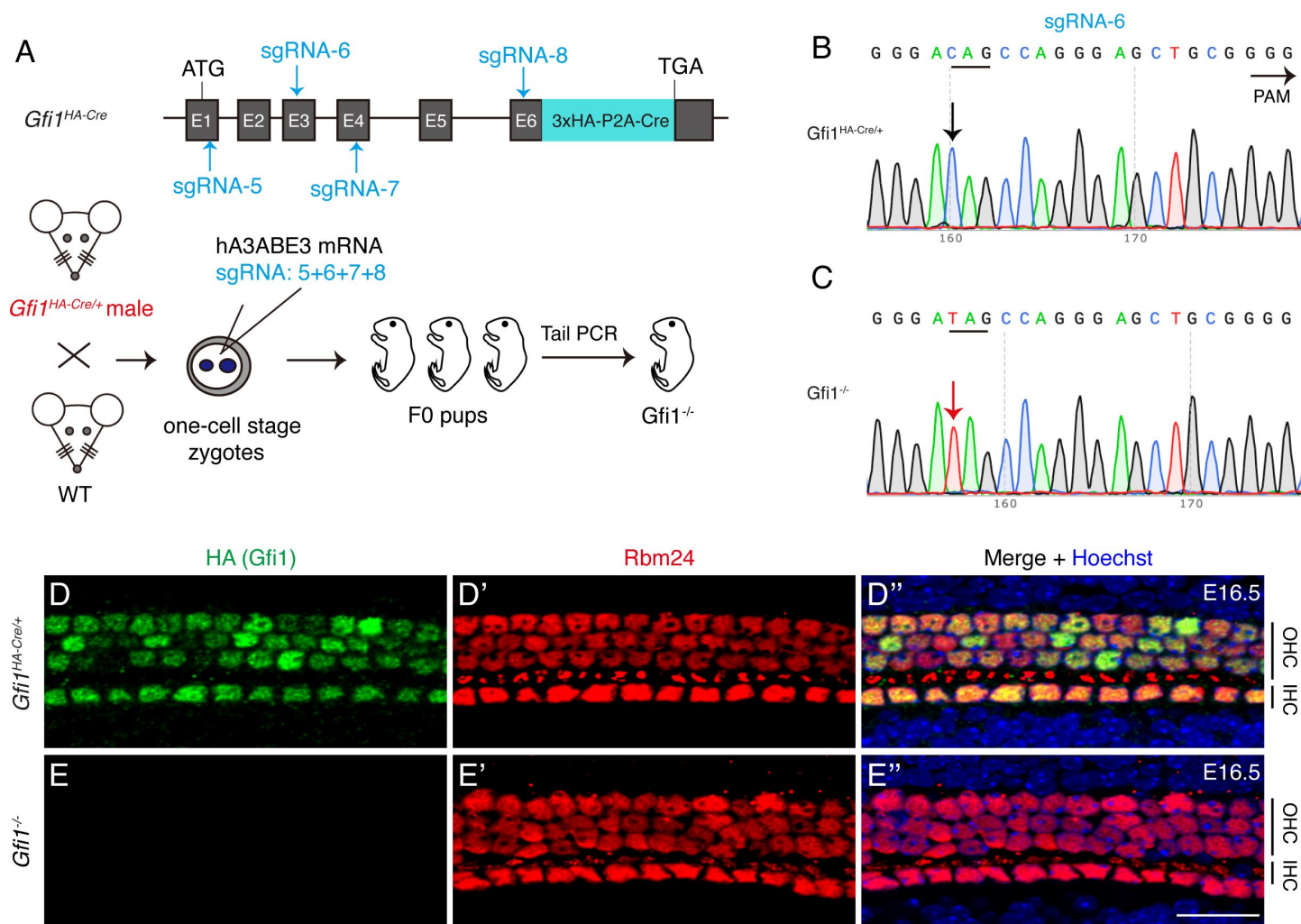


Figure 2.

***Gfi1* is dispensable for *Rbm24* expression.**

(A) A simple cartoon representing how the *Gfi1* was inactivated in the one-cell stage mouse zygotes derived from male *Gfi1*^{HA-Cre/+} by the CRISPR-stop approach. **(B-C)** With sgRNA-6 as an example, the Sanger sequencing chromatograms of control *Gfi1*^{HA-Cre/+} (B) and F0 *Gfi1*^{-/-} (C) were presented. The base 'C' (black arrow in B) was converted in 'T' (red arrow in C), resulting in pre-emergence of protein translation stop codon TAG. The red 'T' was a single peak, meaning that the pre-emergence of stop codon occurred in both alleles. **(D-E'')** Dual staining of HA (Gfi1) and Rbm24 in cochleae from control *Gfi1*^{HA-Cre/+} (D-D'', n=3) and F0 *Gfi1*^{-/-} (E-E'', n=4) at E16.5. Rbm24 expression (E') was not altered in the absence of the HA (Gfi1) expression (E). OHC: outer hair cell; IHC: inner hair cell. Scale bar: 20 μ m.

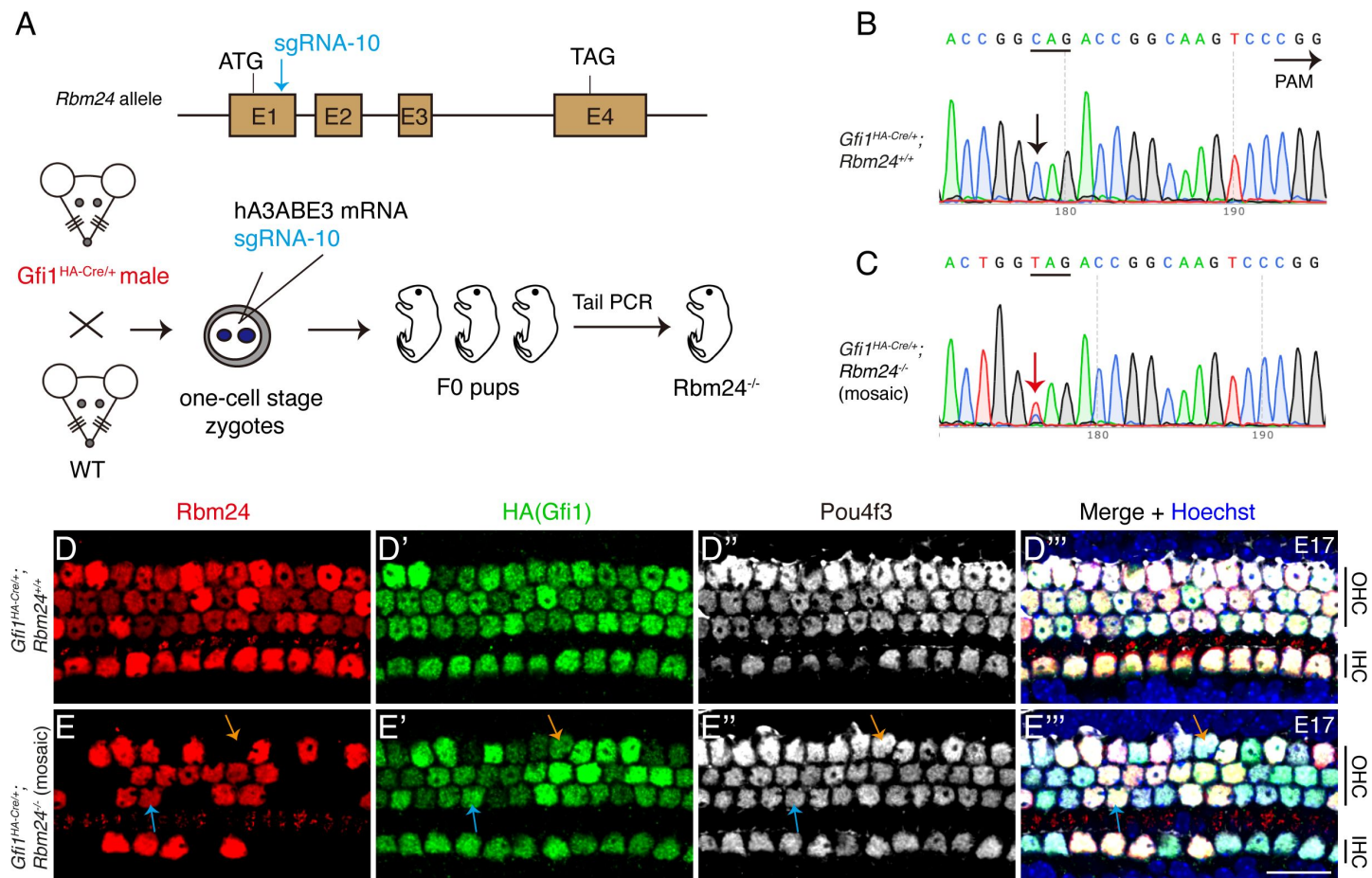


Figure 3.

Pou4f3 and Gfi1 expressions are normal in the absence of *Rbm24*.

(A) A simple cartoon to elaborate how we used CRISPR-stop approach to induce mosaic inactivation of *Rbm24* in the one-cell stage mouse zygote harboring *Gfi1*^{HA-Cre/+}. Notably, the injection was in the late one-cell stage of the zygotes, resulting in higher rates of final mosaic inactivation. **(B-C)** The Sanger sequencing chromatograms of control *Gfi1*^{HA-Cre/+}; *Rbm24*^{+/+} (B) and *Gfi1*^{HA-Cre/+}; *Rbm24*^{-/-} (mosaic) (C). The base 'C' (black arrow in B) was converted into 'T' (red arrow in C) in a fraction of cochlear cells. Notably, the mosaic inactivation of *Rbm24* was evidenced by the double peaks of 'T' and 'C' (red arrow in C). **(D-E''')** Triple staining of *Rbm24*, HA and Pou4f3 in cochlear samples (basal turns) of control *Gfi1*^{HA-Cre/+}; *Rbm24*^{+/+} (D-D''', n=3) and *Gfi1*^{HA-Cre/+}; *Rbm24*^{-/-} (mosaic) (E-E''', n=3) at E17. Consistent with the Sanger sequencing results (C), *Rbm24* was inactivated in cochlear HCs in a mosaic pattern (E). The orange arrows (E-E''') pointed to one OHC that lost *Rbm24*, whereas the blue arrows (E-E''') marked one OHC in which the *Rbm24* expression was maintained. Notably, regardless of whether *Rbm24* was inactivated or not, HA (*Gfi1*) and Pou4f3 expression were not altered. OHC: outer hair cell; IHC: inner hair cell. Scale bar: 20 μm (E''').

Dual staining of HA (Gfi1) and pan-HC marker Myo7a demonstrated that, similar to *Gfi1*^{+/+} mice (n=3, **Supplemental Figure 3E-E**”), HC development in *Gfi1*^{HA-Cre/HA-Cre} was normal at P1 (n=3, **Supplemental Figure 3F-F**”). Moreover, HA (Gfi1) was highly expressed in all Myo7a+ IHCs and OHCs in *Gfi1*^{HA-Cre/HA-Cre} mice (arrows in **Supplemental Figure 3F-F**”). It confirmed that Gfi1 endogenous expression is not affected in the *Gfi1*^{HA-Cre/HA-Cre} mice, representing an advantage over the previous *Gfi1*^{Cre/+} strain where one copy of *Gfi1* is lost and early onset of hearing loss happens in *Gfi1*^{Cre/+} [38](#), [39](#).

We additionally performed the fate mapping analysis of *Gfi1*^{HA-Cre/+}; *Rosa26-loxp-stop-loxp-tdTomato* (Ai9)/+ (*Gfi1*^{HA-Cre/+}; Ai9/+) at P2. The *Gfi1*^{HA-Cre/+}; Ai9/+ model allowed us to both visualize temporal Gfi1 protein expression pattern and permanently mark cells experiencing *Gfi1* expression with tdTomato. Neither tdTomato nor HA was detected in the Myo7a+ HCs of the control Ai9/+ mice (n=5, **Supplemental Figure 3G-G**” and **I-I**”). In contrast, majority of the Myo7a+ HCs expressed tdTomato and HA in the *Gfi1*^{HA-Cre/+}; Ai9/+ mice (n=5, **Supplemental Figure 3H-H**” and **J-J**”). We noticed that the general HA (Gfi1) expression level in IHCs seemed higher than that in OHCs (**Supplemental Figure 3J-J**”). Moreover, HA (Gfi1) expression levels exhibited manifest heterogeneities among OHCs. The OHCs expressing the highest (#1), intermediate (#2) and lowest (#3) levels of HA (Gfi1) were respectively marked (arrows in **Supplemental Figure 3J-J**”). Notably, such a heterogenous Gfi1 expression in neonatal HCs was not observed in *Gfi1*^{HA-Cre/HA-Cre} mice (**Supplemental Figure 3F-F**”), likely due to the higher level of HA tagged Gfi1 in *Gfi1*^{HA-Cre/HA-Cre} than in *Gfi1*^{HA-Cre/+} mice. Besides HCs, tdTomato+ cells were present in the non-sensory regions (arrows in **Supplemental Figure 3H**”), consistent with the previous report [38](#). Collectively, *Gfi1*^{HA-Cre/+} strain was suitable to visualize Gfi1 protein by HA antibody, albeit random insertion of the targeting vector existed. Alternatively, *Gfi1*^{HA-Cre/+} at least can be treated as a pseudo transgenic mouse strain, by which we can reliably visualize Gfi1 and trace cells experiencing *Gfi1* expression.

Gfi1 expression is prohibited in the *Pou4f3*^{-/-} cochlear HCs

Given HA can faithfully represent the Gfi1 expression in the *Gfi1*^{HA-Cre/+}, HA expression was expected to be repressed in the *Pou4f3*^{-/-} HCs, as reported by previous studies [33](#), [34](#). We confirmed this predication by our CRISPR-stop approach [37](#). Basically, the pipeline was identical to the production of *Pou4f3* mutants described above (**Supplemental Figure 1**”), except that here the zygotes were derived from male *Gfi1*^{HA-Cre/+} mice (**Supplemental Figure 4A**”). We obtained the mosaic (or chimeric) *Pou4f3*^{-/-} mice, which was confirmed by Sanger sequencing of tail PCR (**Supplemental Figure 4B**” and **C**”), partly due to the injection time was the late stage of one-cell stage zygotes. In control *Gfi1*^{HA-Cre/+} mice, there was a pure ‘C’ base (blue arrow in **Supplemental Figure 4B**”), however, a mixture of ‘C’ and ‘T’ double peaks existed in *Gfi1*^{HA-Cre/+}; *Pou4f3*^{-/-} (mosaic) mice (red arrow in **Supplemental Figure 4C**”). Nonetheless, the advantage of the mosaic *Pou4f3*^{-/-} mice was that it allowed us to have internal control *Pou4f3*^{+/+} and *Pou4f3*^{-/-} HCs in the same cochlea, and to determine the phenotype occurs in a cell autonomous or non-cell autonomous manner.

In control *Gfi1*^{HA-Cre/+} mice (n=3) at E16.5, all HCs expressed Pou4f3, Rbm24 and HA(Gfi1), albeit HA (Gfi1) levels again seemed heterogenous among the OHCs (**Supplemental Figure 4D-D**”). In contrast, in the *Gfi1*^{HA-Cre/+}; *Pou4f3*^{-/-} (mosaic) mice (n=3), we captured HCs that lost or maintained Pou4f3 expression (**Supplemental Figure 4E-E**”). Notably, Pou4f3+ HCs expressed HA(Gfi1) and Rbm24 (blue arrows in **Supplemental Figure 4E-E**”), in contrast, both HA(Gfi1) and Rbm24 disappeared in HCs that lost Pou4f3 expression (orange arrows in **Supplemental Figure 4E-E**”). Thus, our data clearly confirmed that HA indeed was a reliable readout for Gfi1 expression and was sensitive to loss of Pou4f3. Moreover, it supported that Pou4f3 regulates *Rbm24* in a cell autonomous manner.

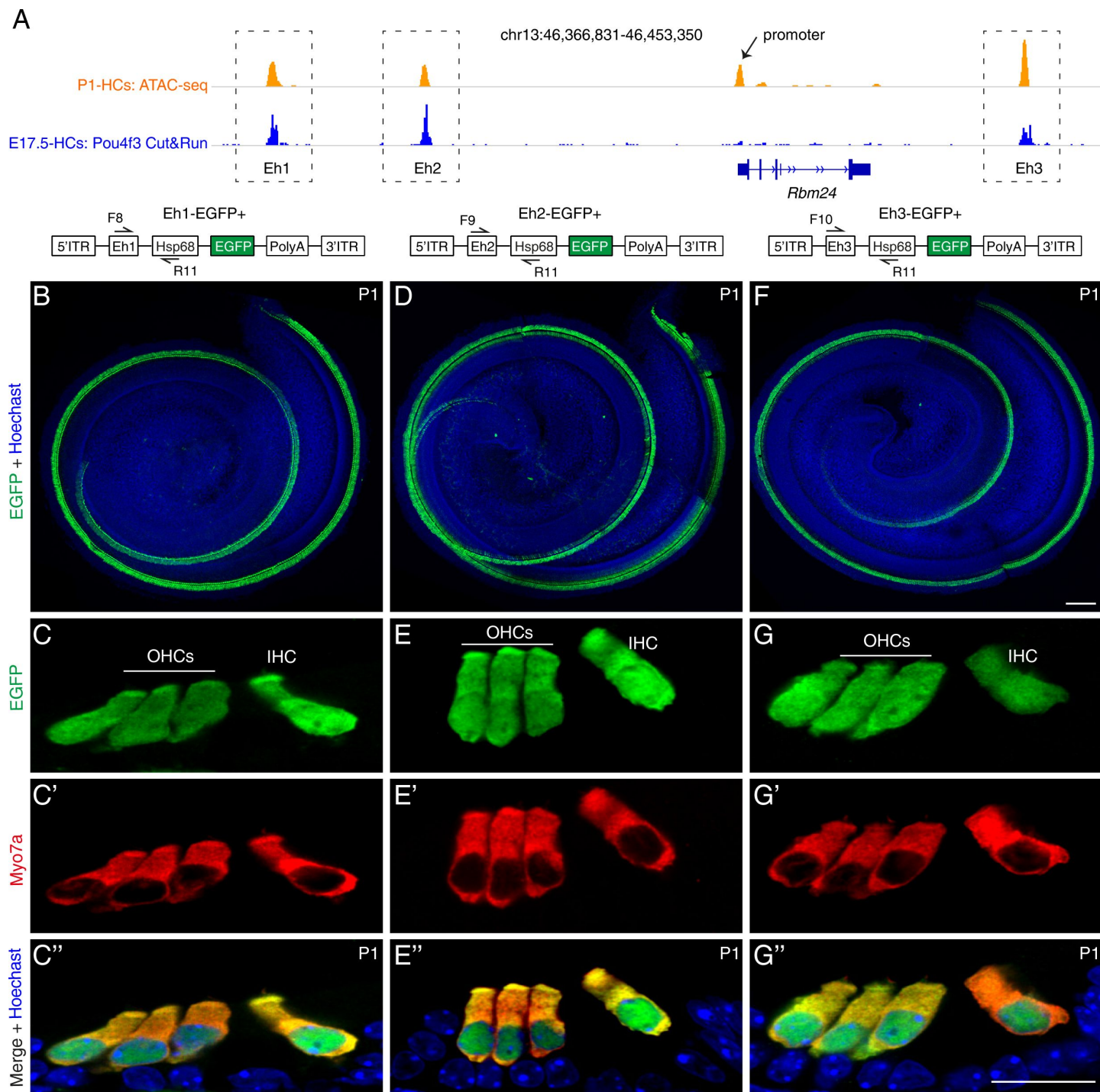


Figure 4.

Three *Rbm24* enhancers are sufficient to drive specific EGFP expression in cochlear HCs.

(A) Visualization of ATAC-seq (top row, orange color) of P1 cochlear HCs (from our previous study, GSE181311) and Pou4f3 Cut&Run assay (bottom row, blue color) of E17.5 cochlear HCs (from another study, GSE150391) by Integrative Genomics Viewer. The black arrow marked the *Rbm24* promoter, and the three *Rbm24* enhancers (Eh1, Eh2 and Eh3) were marked by the dotted square. (B-G'') Dual staining of EGFP and Myo7a in Eh1-EGFP+ (B-C'', n=7), Eh2-EGFP+ (D-E'', n=6) and Eh3-EGFP+ (F-G'', n=4) at P1. Only the EGFP channel was visualized in whole mount staining (B, D, F). In the cryosection cochlear samples, EGFP was overlapped with HC marker Myo7a. OHCs: outer hair cells; IHC: inner hair cell. Scale bars: 100 μ m (F), 20 μ m (G'').

Gfi1 is dispensable for Rbm24 expression

Next, we produced *Gfi1*^{-/-} mutants by the same CRISPR-stop approach³⁷. One-cell stage zygotes derived from male *Gfi1*^{HA-Cre/+} mice were injected with base editor components and 4 different sgRNAs located in different exons of *Gfi1* (**Figure 2A**). The reason why 4 sgRNAs were combined was elaborated in the discussion part. With sgRNA-6 as an example, Sanger sequencing of tail DNA demonstrated that, relative to control *Gfi1*^{HA-Cre/+} mice (black arrow in **Figure 2B**), premature emergence of TAG stop codon happened in F0 mice, resulting in homozygous *Gfi1* inactivation (red arrow in **Figure 2C**). All HCs expressed HA (Gfi1) and Rbm24 in control *Gfi1*^{HA-Cre/+} mice (n=3, **Figure 2D-D**). Notably, Rbm24 expression was maintained in the *Gfi1*^{-/-} HCs that lost HA (Gfi1) at E16.5 (n=4, **Figure 2E-E**). It suggested that Rbm24 expression is not dependent on Gfi1.

The advantage of the above *Gfi1*^{-/-} model was rapid and direct confirming the absence of Gfi1 expression in the F0 mice. However, we cannot guarantee that all HCs completely lost Gfi1 expression at the single cell resolution via HA staining, because only one allele of Gfi1 was *Gfi1*^{HA-Cre}. Thus, to rule out the possibility that only the HA tagged Gfi1 allele was mutated, we further generated the germ line stable *Gfi1* mutants in which its majority DNA fragments between sgRNA-4 and sgRNA-9 were deleted (**Supplemental Figure 5A**). The WT (*Gfi1*^{+/+}), *Gfi1*^{+/-} and *Gfi1*^{-/-} mice were readily to be identified by tail PCR (**Supplemental Figure 5B**). Consistent with previous reports⁴⁰, dual staining of Rbm24 and Pou4f3 showed that, relative to WT mice (n=3, **Supplemental Figure 5C-C**), severe degeneration of HCs, especially of the OHCs, was present in the *Gfi1*^{-/-} mice (n=3, **Supplemental Figure 5D-D**). It validated the success of generating the *Gfi1*^{-/-} mice. Notably, the surviving Pou4f3⁺ HCs kept the expression of Rbm24 (orange arrows in **Supplemental Figure 5D-D**). In contrast, no significant difference was observed between WT and *Gfi1*^{-/-} mice at E16.5, as both WT and *Gfi1*^{-/-} HCs expressed Pou4f3 and Rbm24, and both WT and *Gfi1*^{-/-} OHCs expressed Bcl11b (**Supplemental Figure 5E-F**). It suggests that degeneration of *Gfi1*^{-/-} HCs does not begin by E16.5. Presence of Pou4f3 in *Gfi1*^{-/-} HCs agreed with previous notion that Pou4f3 is epistatic to Gfi1. Collectively, analysis of both *Gfi1* mutant models supports that Gfi1 is dispensable for Rbm24 expression.

Pou4f3 and Gfi1 expressions are normal in *Rbm24*^{-/-} cochlear HCs

Given that Pou4f3 is upstream and positive regulator of *Rbm24*, we predicted that Pou4f3 expression should be unaffected in the *Rbm24*^{-/-} HCs. To determine whether the predication was correct or not, we established the *Rbm24*^{-/-} mouse strain by injecting one *Rbm24* sgRNA (sgRNA-10) into one-cell stage zygotes derived from male *Gfi1*^{HA-Cre/+} mice (**Figure 3A**). We chose *Gfi1*^{HA-Cre/+} zygotes in order to simultaneously assess Gfi1 and Pou4f3 expression patterns in *Rbm24*^{-/-} HCs. Relative to the control *Gfi1*^{HA-Cre/+}; *Rbm24*^{+/+} (**Figure 3B**), premature emerge of the TAG stop codon appeared in the *Gfi1*^{HA-Cre/+}; *Rbm24*^{-/-} (mosaic) mice (**Figure 3C**). Notably, Sanger sequencing of tail DNA showed that the Rbm24 mutation was mosaic, as mixed 'T' and 'C' peak existed (red arrow in **Figure 3C**).

The mosaic inactivation of Rbm24 was further confirmed by triple staining of Rbm24, HA and Pou4f3 (**Figure 3D-E**). Relative to control *Gfi1*^{HA-Cre/+}; *Rbm24*^{+/+} (n=3, **Figure 3D-D**), Rbm24 expression disappeared in a fraction of cochlear HCs (orange arrows in **Figure 3E-E**) but remained normal in other HCs (blue arrows in **Figure 3E-E**) of the *Gfi1*^{HA-Cre/+}; *Rbm24*^{-/-} (mosaic) mice (n=3) at E17. Notably, regardless of whether Rbm24 was inactivated or not in the cochlear HCs, both HA (Gfi1) and Pou4f3 expression patterns were intact. It supports that Pou4f3 is epistatic to Rbm24 and deletion of Rbm24 does not affect Pou4f3 expression. Moreover, our data suggests that Gfi1 and Rbm24 do not have genetic interactions. Gfi1 inactivation did not affect Rbm24 (**Figure 2** and **Supplemental Figure 5**), and vice versa (**Figure 3E-E**).

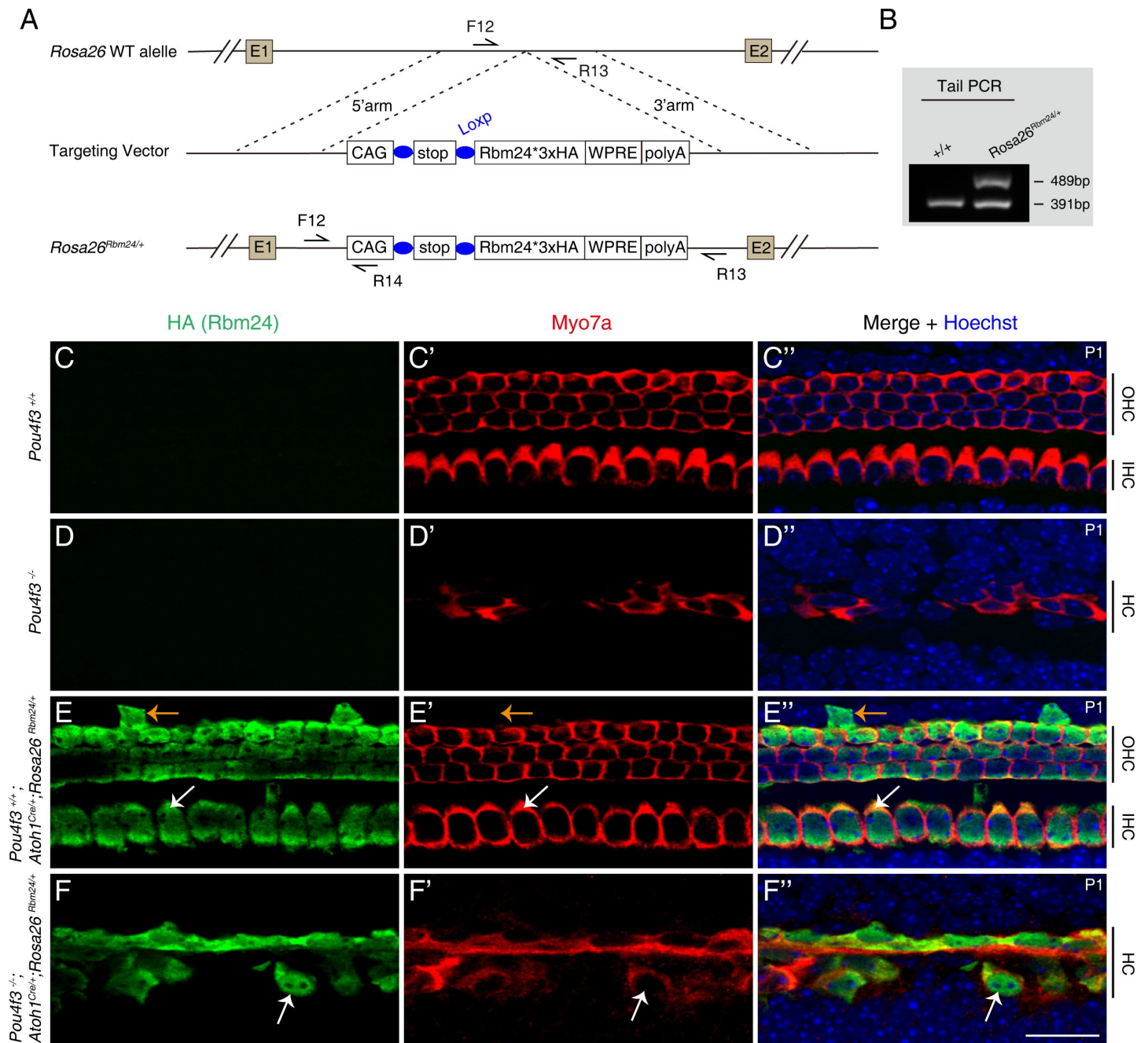


Figure 5.

Forced expression of *Rbm24* fails to rescue the OHC degeneration in *Pou4f3*^{-/-} mutants.

(A) Generation of *Rosa26*-*Loxp-stop-Loxp-Rbm24**3xHA/+ (*Rosa26*^{Rbm24}+/+) mouse strain. The C-terminus of the ectopic *Rbm24* was tagged with three HA tags. **(B)** One example tail PCR gel image that helped us to distinguish WT (+/+) and KI (*Rosa26*^{Rbm24}+/+) mice. **(C-F'')** Dual staining of HA (Rbm24) and hair cell marker Myo7a in mice with four different genotypes at P1: *Pou4f3*^{+/+} (C-C'', n=3), *Pou4f3*^{-/-} (D-D'', n=3) and *Pou4f3*^{+/+}; *Atoh1*^{Cre/+}; *Rosa26*^{Rbm24}+/+ (E-E'', n=3) and *Pou4f3*^{-/-}; *Atoh1*^{Cre/+}; *Rosa26*^{Rbm24}+/+ (F-F'', n=3). The ectopic HA (Rbm24) was not detected in *Pou4f3*^{+/+} (C) and *Pou4f3*^{-/-} (D), but was detected in both *Pou4f3*^{+/+}; *Atoh1*^{Cre/+}; *Rosa26*^{Rbm24}+/+ (E) and *Pou4f3*^{-/-}; *Atoh1*^{Cre/+}; *Rosa26*^{Rbm24}+/+ (F). The white arrows in (E-E'') pointed to one Myo7a+ IHC that expressed the ectopic HA (Rbm24). The HCs seemed normal by P1, despite the ectopic *Rbm24* expression was induced. The orange arrows in (E-E'') labelled one SC that expressed HA (Rbm24) but not Myo7a. The white arrows in (F-F'') should be one remaining Myo7a+ HC that expressed ectopic *Rbm24*. Overall, many of the *Pou4f3*^{-/-} HCs degenerated, albeit ectopic *Rbm24* was induced. OHC: outer hair cell; IHC: inner hair cell; HC: hair cell. Scale bar: 20 μm (F'').

Three *Rbm24* enhancers can drive specific EGFP expression in cochlear HCs

After showing that Pou4f3, but not Gfi1, is indispensable in mediating *Rbm24* expression, we next determined the mechanism underlying how Pou4f3 controls *Rbm24* expression, in particular, the cis-regulation elements (CREs) of *Rbm24*. According to our previous Transposase-Accessible Chromatin with high-throughput sequencing (ATAC-seq) of neonatal cochlear HCs⁸, four CREs of *Rbm24* were identified: one proximal promoter (arrow in **Figure 4A**) and three distal ones that were potential enhancers and defined as Eh1, Eh2 and Eh3, respectively (dotted square in **Figure 4A**). Eh1 and Eh2 were distributed upstream, whereas Eh3 was downstream of *Rbm24* coding region. Moreover, we reanalyzed the Pou4f3 Cut&Run assay from one previous study³², and found that Pou4f3 binds to Eh1, Eh2 and Eh3, but not to the *Rbm24* promoter (**Figure 4A**).

Whether Eh1, Eh2 and Eh3 are bona-fide *Rbm24* enhancers remain to be determined. We reasoned that, if Eh1, Eh2 and Eh3 were *Rbm24* enhancers, either of them, together with the mini promoter of mouse heat shock protein 68 (Hsp68)^{8,41–43}, would be sufficient to drive specific reporter expression in cochlear HCs. To address this question, we established three transgenic mouse strains: Eh1-EGFP+ (n=7, **Figure 4B-C**), Eh2-EGFP+ (n=6, **Figure 4D-E**) and Eh3-EGFP+ (n=4, **Figure 4F-G**). Briefly, in each strain the EGFP expression was driven by the mini promoter of Hsp68 and Eh1 (or Eh2 or Eh3). It is known that the mini promoter of *Hsp68* alone is not able to drive EGFP expression⁴⁴. In contrast, strong EGFP expression was observed by whole mount analysis in all three transgenic lines (**Figure 4B, D** and **F**). Moreover, dual staining of EGFP and Myo7a in cryosectioned cochleae showed that EGFP was specifically expressed in IHCs and OHCs in all three strains at P1 (**Figure 4C-C**, **E-E** and **G-G**). Collectively, our transgenic assay supports that Eh1, Eh2 and Eh3 are *Rbm24* enhancers. Moreover, Pou4f3 mediates the *Rbm24* expression primarily by binding to *Rbm24* enhancers.

Forced *Rbm24* expression fails to alleviate the degeneration of *Pou4f3*^{-/-} HCs

Whether ectopic *Rbm24* expression can alleviate the degeneration of *Pou4f3*^{-/-} HCs remains unknown, albeit we showed that *Rbm24* is certainly regulated by Pou4f3. Thus, we established a new conditional *Rosa26*-Loxp-stop-Loxp-*Rbm24**3xHA/+ (*Rosa26*^{*Rbm24*} in short) mouse model (**Figure 5A** and **B**). To turn on ectopic *Rbm24* in cochlear HCs, *Rosa26*^{*Rbm24*} was further crossed with the *Atoh1*^{*Cre/+*} that targets majority of cochlear HCs and a fraction of SCs³⁹. Notably, in the *Rosa26*^{*Rbm24*} strain, *Rbm24* was tagged with three HA fragments, which facilitated us to distinguish the endogenous and the ectopic *Rbm24*. In the control *Pou4f3*^{*+/+*} (**Figure 5C-C**) and the germ line stable *Pou4f3*^{*-/-*} (**Figure 5D-D**) mice, the Myo7a+ HCs did not express HA (*Rbm24*), however, almost of all Myo7a+ HCs expressed HA (*Rbm24*) in the *Pou4f3*^{*+/+*}; *Atoh1*^{*Cre/+*}; *Rosa26*^{*Rbm24*} mice at P1 (white arrows in **Figure 5E-E**). As expected, besides HCs, we captured a few SCs expressing HA (*Rbm24*) but not Myo7a (orange arrows in **Figure 5E-E**). Those cochlear HCs expressing HA (*Rbm24*) seemed normal, suggesting that they can tolerate extra *Rbm24* expression by P1. Collectively, it supports that the *Rosa26*^{*Rbm24*} model was successfully generated and ectopic *Rbm24* was induced in cochlear HCs in *Pou4f3*^{*+/+*}; *Atoh1*^{*Cre/+*}; *Rosa26*^{*Rbm24*}.

In contrast to our expectation, ectopic *Rbm24* expression failed to mitigate the degeneration of *Pou4f3*^{*-/-*} HCs. Severe degeneration of HCs (both IHCs and OHCs) occurred in *Pou4f3*^{*-/-*}; *Atoh1*^{*Cre/+*}; *Rosa26*^{*Rbm24*} (**Figure 5F-F**), indistinguishable from the *Pou4f3*^{*-/-*} mice (**Figure 5D-D**), except that the remaining HCs expressed *Rbm24* in the *Pou4f3*^{*-/-*}; *Atoh1*^{*Cre/+*}; *Rosa26*^{*Rbm24*} (arrows in **Figure 5F-F**). It suggests that restoration of *Rbm24* alone is not sufficient to prevent *Pou4f3*^{*-/-*} cochlear HCs from undergoing degeneration.

Molecular mechanisms underlying cochlear HC survival

Our sound receptor HCs are vulnerable to various genetic mutations, environmental ototoxic factors and ageing. The degeneration of HCs is one of the primary reasons accounting for human sensorineural hearing impairment ⁴⁵. Previous studies have revealed many genes whose mutations lead to HC degeneration starting at different ages. Severe HC developmental defect happens in *Atoh1*^{-/-} mutants ^{7, 46}. It is known that *Atoh1*^{-/-} cochlear sensory cells undergo apoptosis. Whether the absence of HCs in *Atoh1*^{-/-} mice is due to that HCs are never produced, or immediately die after initial emergence remains unclear, partly due to the difficulty of unambiguously defining the nascent HCs by molecular markers.

Differently from *Atoh1*^{-/-} mutants, in both *Pou4f3*^{-/-} and *Gfi1*^{-/-} mice ^{31, 33, 34}, the initial cochlear HC development is normal but become defective at perinatal ages, consistent with the notion that Pou4f3 and Gfi1 are not required in HC fate specification but needed in subsequent differentiation and survival. It is known that caspase 3 is active during the hair cell death in the *Pou4f3*^{-/-} mutants, and the anti-apoptotic factor z-VAD-fmk has a protective effect on the *Pou4f3*^{-/-} HCs between 14.5 and E16.5 ⁴⁷. Moreover, we noticed that the overall HC degeneration in *Gfi1*^{-/-} was milder than in *Pou4f3*^{-/-} mice at P1, because only OHC degeneration happened in the *Gfi1*^{-/-}, but both IHC and OHC were degenerated in *Pou4f3*^{-/-} mice (**Supplemental Figures 2** and **5**).

It is known that *Gfi1* is the target of Pou4f3 ³⁴. Whether forced Gfi1 expression can alleviate the degeneration of *Pou4f3*^{-/-} HCs remains unknown, too. Likewise, Pou4f3 is one of the target genes regulated by Atoh1 ³², however, whether forced expression of either Pou4f3 or Gfi1, or both Pou4f3 and Gfi1, is able to mitigate the developmental defects of *Atoh1*^{-/-} HCs remains to be determined. Future studies are warranted to resolve those remaining questions before further deeply understanding the mechanisms underlying how to keep cochlear HCs to survive.

Roles of Rbm24 and how Rbm24 is regulated during cochlear HC development

Shortly after cochlear HCs emerge, Rbm24 is expressed and permanently maintained, evidenced by transcriptomic analyses and antibody staining ^{23, 35, 48–50}. Rbm24 is not necessary in early phase of cochlear HC development, because *Rbm24*^{-/-} HC development is normal by P1, however, Rbm24 is required for OHC to survive after birth ³⁶. *Rbm24*^{-/-} OHCs, but not IHCs, are degenerated by P19 ³⁶. Rbm24 is also involved in mRNA stability and pre-mRNA alternative splicing of genes including *Cdh23* and *Pcdh15* that are crucial for HC stereocilia development ^{51–53}. Nonetheless, the single-cell transcriptomic analysis of *Rbm24*^{-/-} IHCs or OHCs is not yet available and the detailed molecular mechanism underlying *Rbm24*^{-/-} HC degeneration remains to be clarified in future studies.

What are the *trans*-acting factors that are involved in regulating *Rbm24* expression? First, our current study provided strong genetic evidence to support that Pou4f3 is needed in turning on *Rbm24*. In the absence of Pou4f3, Rbm24 expression cannot be triggered. The normal expression of Pou4f3 in *Rbm24*^{-/-} HCs further confirmed that *Pou4f3* is epistatic to *Rbm24* and that Rbm24 is dispensable for *Pou4f3* expression. Besides Pou4f3, the engagement of Atoh1 in *Rbm24* regulation is supported by two lines of evidence: one is that *Rbm24* expression disappears in the *Atoh1*^{-/-} cochlear HCs ⁵⁴; the second is that *Rbm24* belongs to the Atoh1 binding targets revealed by Atoh1 Cut&Run assay ^{8, 32}. Notably, Atoh1 binds both *Rbm24* promoter and its three enhancers Eh1, Eh2 and Eh3 ^{8, 32}, which differs with Pou4f3 which only binds *Rbm24* enhancers (**Figure 4**). Thus, it is likely that Pou4f3 and Atoh1 cooperate to regulate *Rbm24*, and

either *Pou4f3* or *Atoh1* mutation leads to repression of *Rbm24* expression. Whether *Rbm24* expression in adult HCs needs *Pou4f3* remains to be determined by future conditional *Pou4f3* loss-of-function studies.

Gfi1 and Rbm24 expression are independent to each other

We initially hypothesized that *Pou4f3* mediated *Rbm24* expression through *Gfi1*. However, this hypothesis is not supported by the observation that *Rbm24* expression is normal in the *Gfi1*^{-/-} HCs (Figure 2 and Supplemental Figure 5). Moreover, in the absence of *Rbm24*, *Gfi1* expression is not altered in cochlear HCs, either. Thus, *Rbm24* and *Gfi1* expressions seem independent to each other. It might be due to the functional difference between *Pou4f3* and *Gfi1* during cochlear HC development. Both *Pou4f3* and *Gfi1* are needed in promoting expression of genes involved in HC differentiation^{28, 30, 34, 40}, however, *Gfi1*, but not *Pou4f3*, also represses the preceding neural genes expressed in nascent HCs⁴⁰.

Furthermore, we have another observation relevant to *Gfi1* isoforms that is worthy of discussion. Our previous report shows that one efficient sgRNA is enough to induce homozygous gene inactivation by CRISPR-stop³⁶. Notably, we used 4 different *Gfi1* sgRNAs that are distributed in different exons (Figure 2A). We indeed successfully established one *Gfi1* mutant by using the sgRNA-5 alone which in principle would effectively pre-stop the *Gfi1* translation in exon 1 coding SNAG repressor domain. This *Gfi1* mutant (sgRNA-5 alone) was able to yield the HC degeneration phenotype, but the HA (*Gfi1*) was still detectable in HCs. It is consistent with the notion that *Gfi1* mutant lacking the SNAG domain is equivalent to *Gfi1*-null model⁵⁵. Thus, it is likely that *Gfi1* has multiple unknown isoforms, many of which might be recognized by the HA tag antibody because HA is tagged in the last exon 6. When 4 *Gfi1* sgRNAs were combined, we were able to both reproduce the HC degeneration phenotype and completely block the HA (*Gfi1*) expression.

Why does ectopic Rbm24 fail to alleviate

the degeneration of *Pou4f3*^{-/-} HCs?

After confirming the epistatic genetic interaction between *Pou4f3* and *Rbm24*, we predicted that ectopic *Rbm24* expression should be able to, at least partly, alleviate the degeneration of *Pou4f3*^{-/-} HCs, similar to the case between *Insm1* and *Ikzf2* in cochlear OHCs in our recent study²¹. However, the actual observation is that *Rbm24* failed to rescue the HC degeneration in the *Pou4f3*^{-/-} mice. There are two possible interpretations: 1) Despite it is regulated by *Pou4f3*, *Rbm24* is not directly involved in the pathway regulating HC survival. Indeed, the cell death in *Rbm24*^{-/-} HCs happens later than in *Pou4f3*^{-/-} HCs; 2) Although *Rbm24* is a key *Pou4f3* downstream target, its forced expression alone cannot compensate loss of *Pou4f3* or other known *Pou4f3* targets such as orphan thyroid nuclear receptor *Nr2f2* and *Caprin-1*^{56, 57}. *Caprin-1* is recruited to stress granules in cochlear HCs exposed to ototoxic trauma⁵⁷.

The potential application of the three *Rbm24* enhancers in cochlear HC gene therapy

Gene therapy is a promising strategy to restore the hearing capacity in human inheriting gene mutations⁴⁵. A few examples have been reported, including *Otoferlin* and *vGlut3* gene replacement^{58–61}. Currently, the therapeutic cDNAs were primarily delivered into HCs with an adeno-associated virus (AAV) vector. Despite many AAVs are reported^{62, 63}, their transfecting cochlear cells is non-selective and HC specific AAV is not available yet. One solution is that, instead of the CAG/CMV ubiquitous promoter widely used in current AAVs^{64, 65}, either of the three *Rbm24* enhancers together with *Hsp68* mini promoter (in brief *Rbm24-Hsp68*) should be able to produce an AAV that would specifically transfect HCs. Moreover, *Rbm24* is permanently

expressed in cochlear HCs. Given the *Rbm24-Hsp68* AAV works as expected, it would be a powerful tool for future HC specific gene therapy at all postnatal ages, with potential applications in treating clinical human deafness.

Methods and materials

Mice

The *Atoh1*^{Cre/+} model was kindly provided by Dr Lin Gan (Augusta University, USA)³⁹. The *Rosa26-loxp-stop-loxp-tdTomato* (Ai9)/+ strain (Jax#: 007905) was from the Jackson Laboratory. All mice were bred and raised in a SPF-level animal room and all animal procedures were performed according to the guidelines (NA-032-2022) of the IACUC of the Institute of Neuroscience (ION), Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences.

One-step generation of homozygous *Pou4f3* or *Gfi1* or *Rbm24* mutants by CRISPR-stop

The detailed protocol of how to use CRISPR-stop to generate homozygous gene mutants is described in our previous report^{36, 37}. Briefly, efficient pre-tested sgRNAs (Table S1) and hA3ABE3 were co-injected into one-cell stage mouse zygotes that were subsequently transplanted into pseudopregnant female mice giving birth to the founder 0 (F0) mice. With Sanger sequencing of tail PCR, the F0 mice with homozygous mutation (pre-emergence of protein translation stop codon) were identified and immediately ready for analysis.

Generating *Pou4f3* or *Gfi1*-null mutants with large DNA fragment deletion by CRISPR/Cas9

To construct germ line stable null mutants of either *Pou4f3* or *Gfi1*, *Cas9* mRNA, two efficient pre-tested sgRNAs respectively located at the proximal and distal end of the targeted gene, together with a single strand DNA donor (120 bp) (Table S3), were co-injected into the one-cell stage wild type zygotes. Notably, the left half (60 bp) of the single strand DNA donor was homologous to the 5' end, and the right half (60 bp) was to the 3' end of the targeted gene. The post-injected zygotes were transplanted into pseudopregnant females that would give birth to the F0 mice subjected to tail PCR screening (primers were listed in Table S2). The F0 mice harboring the designed large DNA deletion between the two sgRNAs were identified and further bred with wild type mice to establish the germ-line stable mutants (F1 or afterward).

Construction of the *Gfi1**3×HA-P2A-Cre knockin mouse strain

The sgRNA against *Gfi1* (5'-ATGGACTCAAATGAGTACCC-3'), *Cas9* mRNA and the targeting vector (Supplemental Figure 3B) were co-injected into the one-cell stage wild type mouse zygotes. The targeting vector contained three portions, 5' homologous arm (800 bp), 3' homologous arm (800 bp) and the part in the middle of 5' and 3' arms that contained three HA fragments followed by 2A-Cre. The F0 mice with the potential gene targeting (Supplemental Figure 3C) were screened by tail PCR, followed by crossing with wild type mice to produce F1 mice. Those F1 mice were confirmed by tail PCR again, and further subjected to Southern blot. The detailed protocol of Southern blot is described in our previous report⁶⁶. Tail DNA PCR was used for routine genotyping and primers used to distinguish between KI (*Gfi1*^{HA-Cre/+}) and WT alleles were F4, R5 and R6 (Table S2).

Construction of Eh1-EGFP+, Eh2-EGFP+ and Eh3-EGFP+ transgenic reporter lines

All three transgenic reporter mouse lines, Eh1-EGFP+, Eh2-EGFP+ and Eh3-EGFP+ were produced by the same experimental procedures. The core DNA sequences of each enhancer (Eh1 or Eh2 or Eh3 in [Figure 4A](#)), mRNA of PiggyBac transposase, and the PiggyBac vector ([Figure 4B](#), [D](#) and [F](#)) were co-injected into the one-cell stage of wild type mouse zygotes. The PiggyBac vector contained Eh1/Eh2/Eh3 core DNA sequences (Table S3), mini promoter of mouse heat shock protein 68 (Hsp68) and the EGFP coding sequence. The PiggyBac vector was randomly integrated into mouse genome by the transposase. The F0 mice containing the PiggyBac vector were screened by tail PCR and further bred with wild type mice to produce germ line stable F1 transgenic reporter strains. The samples analyzed in [Figure 4B-G](#) were from F1 or afterwards. Primers used for genotyping transgenic reporter strain were F8, F9 and F10 and a common primer R11 and primer sequences were listed in Table S2.

Generation of *Rosa26*^{CAG-*Isl-Rbm24**3×HA/+} mouse models

The *Rosa26*^{CAG-*Isl-Rbm24**3×HA/+} (*Rosa26*^{*Rbm24*+/+}) knockin mouse strain was constructed by homologous recombination mediated by CRISPR/Cas9. The pre-tested *Rosa26* sgRNA (5'-*ACTCCAGTCTTTCTAGAAGA*-3'), targeting vector ([Figure 5A](#)), together with the mRNA of *Cas9* were co-injected into the one-cell stage wild type mouse zygotes. Similarly, F0 mice with potentially correct gene targeting were screened by tail PCR and further bred with wild type mice to establish germ line stable F1 mice. The primers used to distinguish wild type and the KI (*Rosa26*^{*Rbm24*+/+}) alleles were F12, R13 and R14 (Table S2).

Sample processing and immunofluorescence assay

Inner ears were dissected and fixed in 4% paraformaldehyde (PFA) in PBS (E607016-0500, Sangon Biotech) at 4°C overnight. For cochlear cryosections, inner ears were dehydrated in 30% sucrose (V900116, Sigma) at 4°C before embedding into the optimal cutting temperature (OCT) compound (4583, SAKURA) and slices were cut with 14 µm thickness. The detailed protocol of immunofluorescent staining is described in our previous report [67](#). The following primary antibodies were used: anti-Rbm24 (rabbit, 1:500, 18178-1-AP, Proteintech); anti-Pou4f3 (mouse, 1:500, sc-81980, Santa Cruz); anti-Pou4f3 (rabbit, 1:500, NBP1-88349, Novus Biologicals); anti-HA (rat, 1:500, 11867423001, Roche); anti-Myo7a (rabbit, 1:500, 25-6790, Proteus Biosciences); anti-Myo7a (mouse, 1:500, MYO7A 138-1, Developmental Studies Hybridoma Bank); anti-Ctip2 (Bcl11b) (rat, 1:500, ab18465, Abcam); anti-Insm1 (guinea pig, 1:6000, a kind gift from Dr. Carmen Birchmeier from Max Delbrueck Center for Molecular Medicine, Germany); anti-GFP (chicken, 1:500, ab13970, Abcam). Different corresponding Alexa Fluor-conjugated secondary antibodies were used for detecting the primary antibodies above. Hoechst 33342 (1:1000, H3570, Thermo Fisher Scientific) was used for nuclear DNA staining.

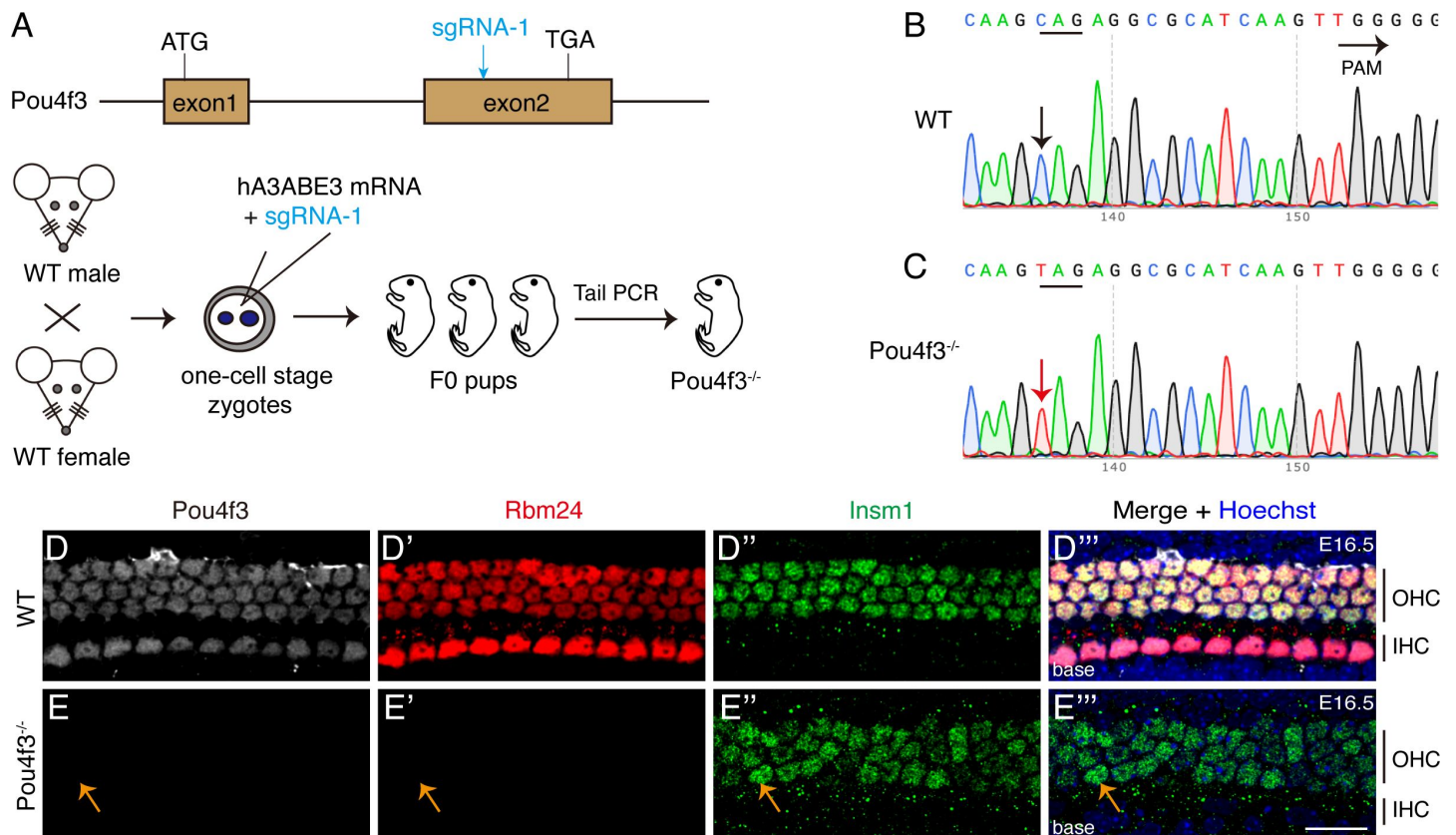
Lastly, either whole-mount or cryosection samples were mounted with Prolong Gold antifade medium (P36930, Thermo Fisher Scientific) at room temperature for 12 h. Samples were scanned by the Nikon C2 or Nikon NiE-A1 Plus confocal microscope. ImageJ software was used to process the confocal images.

Cell quantification and statistical analysis

Before immunofluorescent staining, each cochlear sample was grossly separated into three portions but with different length. Each portion of the same cochlea was initially scanned with confocal microscope but at a low magnification (10×lens). After calculating the total length of each cochlea, it was precisely divided into basal, middle and apical turn with equal length. Then, for cell counting in the experiments ([Supplemental Figure 2D](#)), in each turn ~200µm length of the sensory epithelium was scanned with confocal microscope but at a high magnification (60×lens)

by which the number of HCs was counted. In the *Pou4f3*^{-/-} cochleae, the percentage of the surviving HCs was calculated by normalizing the number of the remaining HCs to the counterparts in the wild type mice. All cell numbers were presented as means $\pm$ SEM. For statistical analyses, we used GraphPad Prism 6.0 software and performed Student's *t* tests with Bonferroni correction.

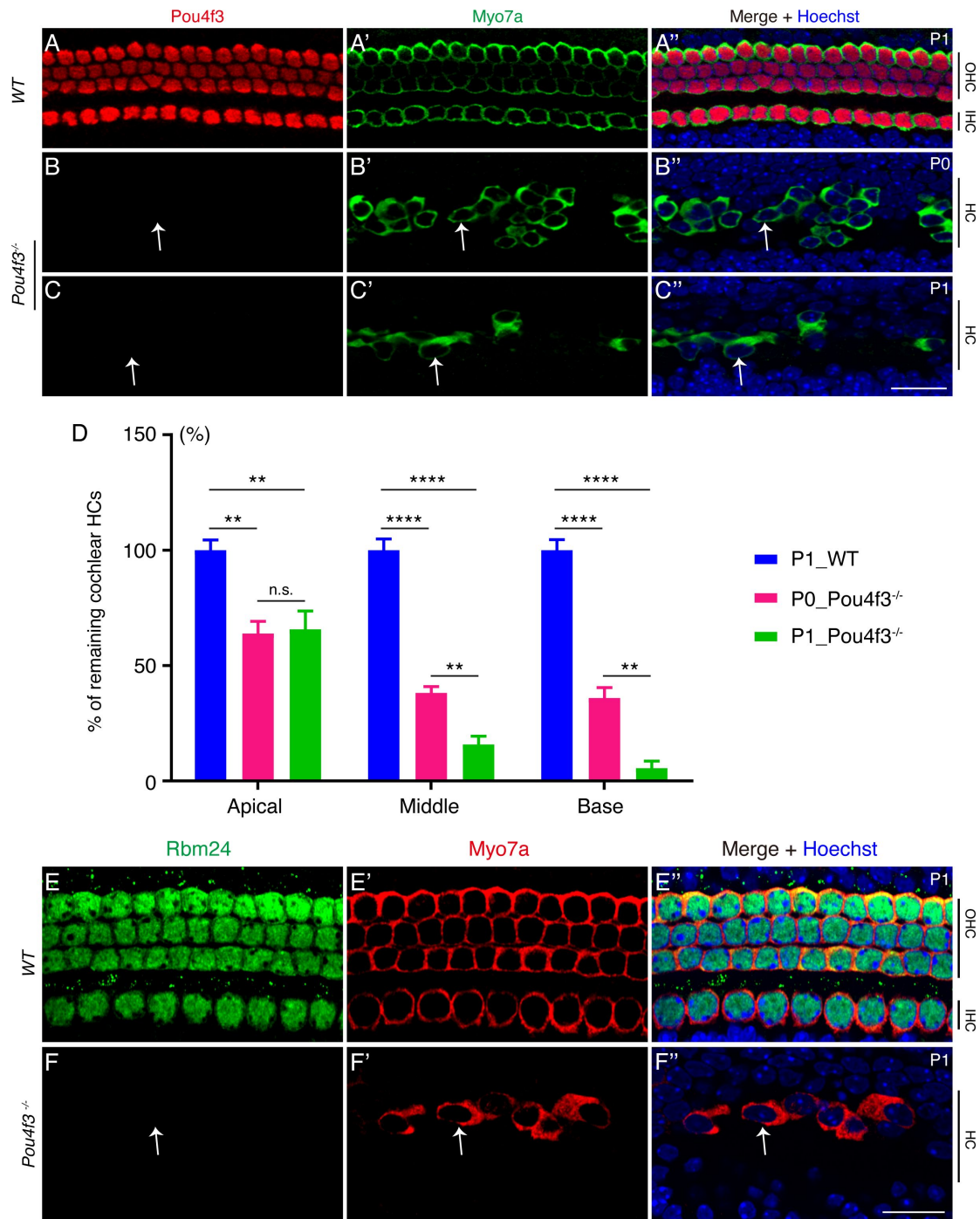
Supplemental Figure Legends



Supplemental Figure 1.

Rbm24 expression is repressed in cochlear HCs at E16.5 when Pou4f3 is inactivated.

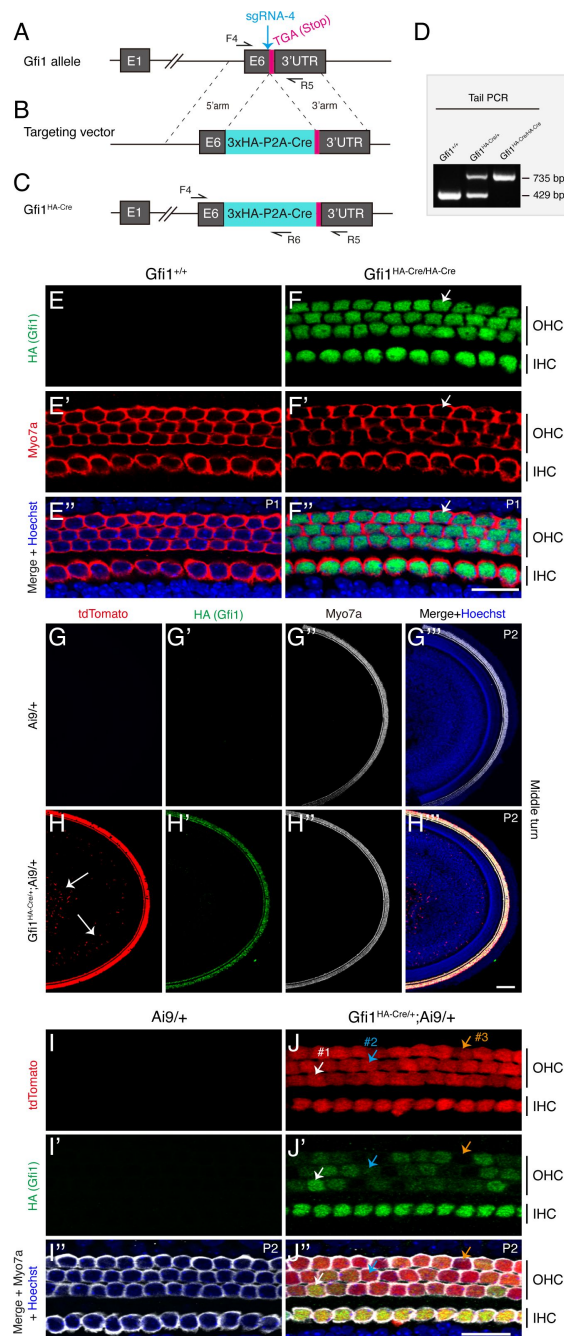
(A) An illustration of how *Pou4f3*^{-/-} mice (Founder 0, F0) were rapidly generated by CRISPR-stop. **(B-C)** Sanger sequencing chromatograms of WT (B) and F0 *Pou4f3*^{-/-} (C) were presented. The base 'C' (black arrow in B) was completely converted to 'T' (red arrow in C), resulting in pre-emergence of protein translation stop codon. The *Pou4f3*^{-/-} mutants (F0 mice) were immediately ready for phenotypic analysis. **(D-E''')** Triple staining of pan-HC marker Pou4f3, Rbm24 and OHC marker Insm1 in whole mount prepared cochlear samples (basal turns) of WT (D-D''', n=3) and *Pou4f3*^{-/-} (E-E''', n=9) at E16.5. Relative to WT mice, both Pou4f3 and Rbm24 expression disappeared in *Pou4f3*^{-/-} HCs, whereas the Insm1 expression was normal. The orange arrows in (E-E''') pointed to one Insm1+ OHC that lost Pou4f3 and Rbm24. OHC: outer hair cell; IHC: inner hair cell. Scale bar: 20 μm (E''').



Supplemental Figure 2.

Severe degeneration of *Pou4f3*^{-/-} HCs happens by perinatal ages.

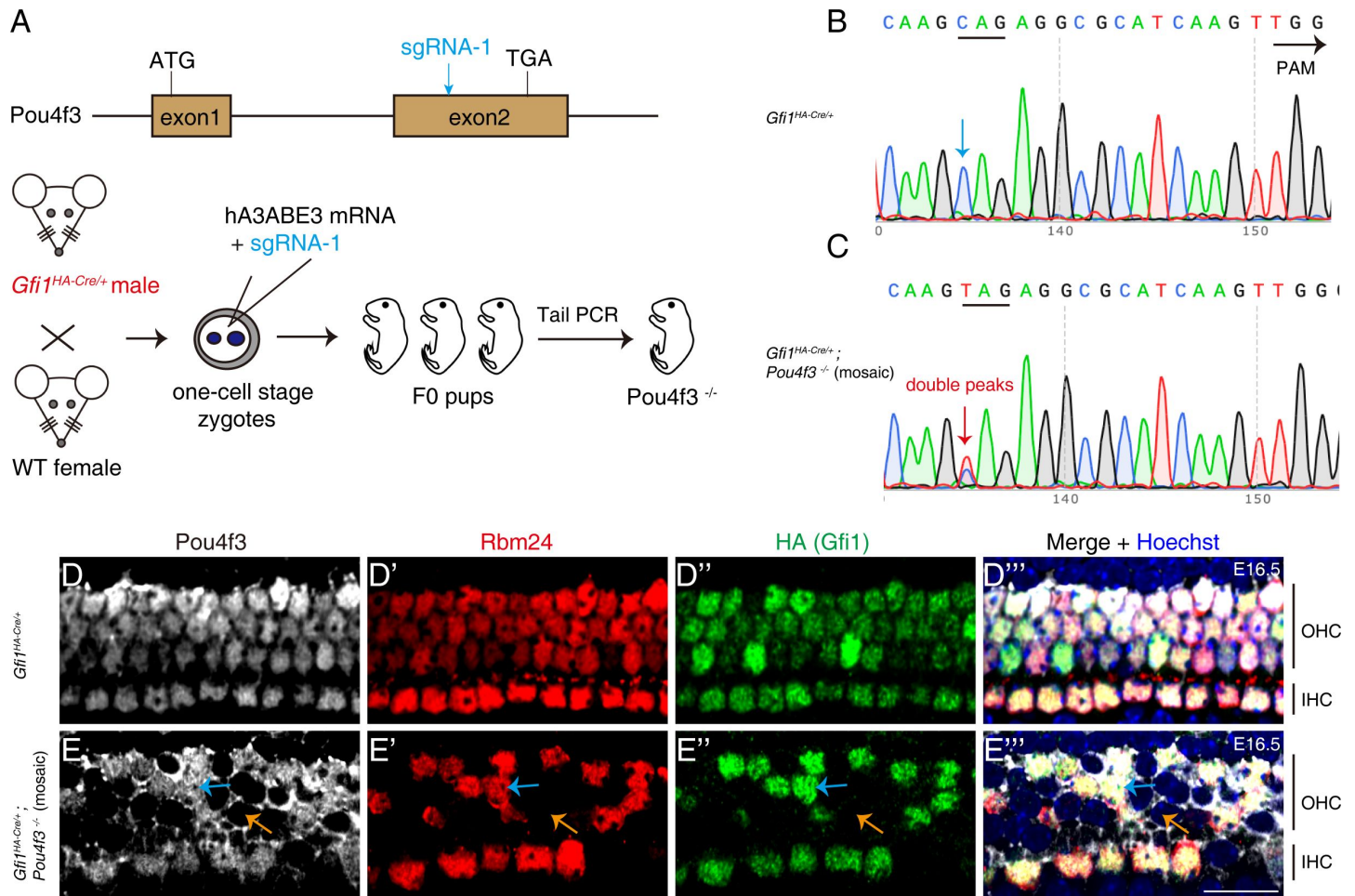
(A-C'') Dual staining of Pou4f3 and Myo7a in middle turn regions of the cochleae of WT (A-A'') at P1 and *Pou4f3*^{-/-} mice at P0 (B-B'') and at P1 (C-C''). Notably, here the *Pou4f3*^{-/-} mice were germ line stable strain produced by CRISPR/Cas9-mediated large DNA fragment deletion (Figure 1A). Arrows in (B-C'') represented the surviving HCs in the absence of Pou4f3. (D) Quantification of the percentage of the remaining HCs at basal, middle, and apical turns in *Pou4f3*^{-/-} mice at P0 (n=4) or at P1 (n=3), relative to WT mice at P1 (n=3). Data were presented as Mean ± Sem. Student's *t* test was used for statistical analysis. ** *p* < 0.01; **** *p* < 0.0001; n.s. no significant difference. HC degeneration at middle and basal turns was more severe at P1 than that at P0, but there was no significant difference in apical turns (D). (E-F'') Dual staining of Rbm24 and Myo7a in WT (E-E'', n=3) and *Pou4f3*^{-/-} (F-F'', n=3) mice at P1. The white arrow in (F-F'') marked one remaining *Pou4f3*^{-/-} HC that lost Rbm24 expression. OHC: outer hair cell; IHC: inner hair cell; HC: hair cell. Scale bar: 20 μm (C'' and F'').



Supplemental Figure 3.

Construction of *Gfi1*^{HA-Cre/+} (*Gfi1*^{HA-Cre/+}) mouse model by which HA can faithfully represent *Gfi1* expression.

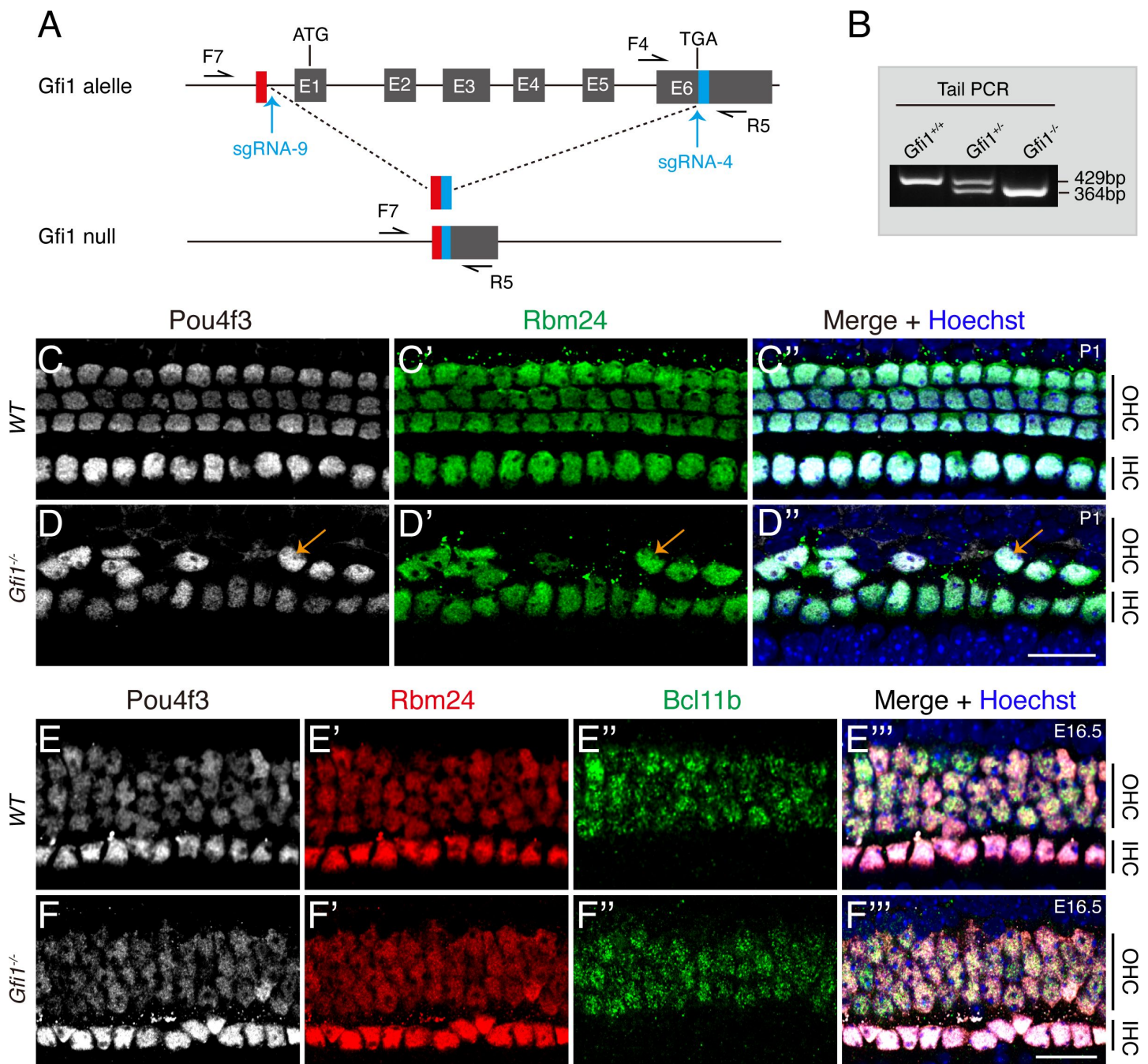
(A-C) A simple cartoon to illustrate how the *Gfi1*^{HA-Cre/+} strain was constructed. Basically, the WT *Gfi1* allele (A) was inserted a fragment containing 3xHA-P2A-Cre (B) immediately before the TGA stop codon (red in A), resulting in a post-targeted *Gfi1*^{HA-Cre/+} allele (C). (D) One example tail PCR gel image that helped us to distinguish *Gfi1*^{+/+}, *Gfi1*^{HA-Cre/+} and *Gfi1*^{HA-Cre/HA-Cre} mice. (E-F'') Dual staining of HA (Gfi1) and HC marker Myo7a of control *Gfi1*^{+/+} (E-E'', n=3) and homozygous *Gfi1*^{HA-Cre/HA-Cre} mice (F-F'', n=3) at P1. Arrows in (F-F'') marked one OHC that expressed HA (Gfi1) and Myo7a. (G-H'') Fate mapping analysis of cells experiencing *Gfi1* expression by *Gfi1*^{HA-Cre/+}; *Rosa26-loxp-stop-loxp-tdTomato* (*Ai9*)/+ (*Gfi1*^{HA-Cre/+}; *Ai9*)/+, with confocal low magnification images being presented in (G-H'') and high magnification images in (I-J''). In *Ai9*+/+ control mice at P2 (n=5), no Tdtomato+ cells were detected (G-G''' and I-I'''). In contrast, in the *Gfi1*^{HA-Cre/+}; *Ai9*+/+ mice (n=5), Tdtomato was detected in both non-sensory cells (white arrows in H) and the Myo7a+ HCs that expressed heterogeneous levels of HA (Gfi1), as represented in different arrows in (J-J''): white arrows (#1) marked one OHC with the highest HA (Gfi1), the orange arrows (#3) with the lowest (or undetectable) HA (Gfi1), and the blue arrows (#2) with the intermediate level of HA (Gfi1). Scale bars: 100µm in (H'''); 20 µm (F'' and J'').



Supplemental Figure 4.

Gfi1 expression is lost in the *Pou4f3*^{-/-} HCs.

(A) One-step generation of *Pou4f3* mutants by CRISPR-stop, with the zygotes derived from the *Gfi1*^{HA-Cre/+} mice. **(B-C)** Sanger sequencing chromatograms of tail DNA from the control *Gfi1*^{HA-Cre/+} (B) and *Gfi1*^{HA-Cre/+}; *Pou4f3*^{-/-} (mosaic) mice (C). The base 'C' in (blue arrow in B) was converted to 'T' (red arrow in C). However, the conversion was not complete because double peak existed, supporting that the inactivation of *Pou4f3* was mosaic. **(D-E''')** Triple staining of Pou4f3, Rbm24 and HA (Gfi1) in the basal region of cochleae of *Gfi1*^{HA-Cre/+} (D-D''', n=3) and *Gfi1*^{HA-Cre/+}; *Pou4f3*^{-/-} (mosaic, E-E''', n=3) at E16.5. Blue arrows in (E-E''') marked one OHC that expressed Pou4f3, Rbm24 and HA (Gfi1), whereas the orange arrows labelled the other OHC that lost Pou4f3, Rbm24 and HA (Gfi1). Thus, it confirmed that not only HA faithfully represented the Gfi1 expression, but also HA expression was dependent on Pou4f3. Scale bar: 20 μ m (E''').



Supplemental Figure 5.

Rbm24 expression is normal in the absence of *Gfi1*.

(A) The DNA fragment between the sgRNA-4 and sgRNA-9 within the *Gfi1* locus was deleted by CRISPR/Cas9 method. **(B)** One example tail PCR gel image that helped us to distinguish *Gfi1*^{+/+}, *Gfi1*^{+/-} and *Gfi1*^{-/-} mice. **(C-D'')** Dual staining of Pou4f3 and Rbm24 in middle turn cochleae of WT (C-C'', n=3) and *Gfi1*^{-/-} (D-D'', n=3) at P1. Severe HC, especially OHC, degeneration happened in *Gfi1*^{-/-} cochleae. The orange arrows in (D-D'') marked one surviving *Gfi1*^{-/-} OHC that expressed Pou4f3 and Rbm24. **(E-F''')** Triple staining of Pou4f3, Rbm24 and OHC marker Bcl11b of cochleae of WT (E-E'', n=3) and *Gfi1*^{-/-} (F-F'', n=3) at E16.5. In terms of the expression patterns of Pou4f3, Rbm24 and Bcl11b, no significant difference was captured between WT and *Gfi1*^{-/-} cochleae at E16.5, suggesting that *Gfi1*^{-/-} HC degeneration did not start yet. Scale bars: 20 μ m (D'' and F''').

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Reviewer #1 (Public Review):

Wang and colleagues recently demonstrated the essential role of RBM24 (RNA-binding motif protein 24a) in the development of mouse hair cells (source: <https://doi.org/10.1002/jcp.31003>). In this study, they further expand on their findings by revealing that Rbm24 expression is absent in Pou4f3 mutant mice but not in Gfi1 mutant mice. This observation suggests that POU4F3 acts as an upstream regulator of Rbm24. The researchers effectively demonstrate that POU4F3 can bind to and regulate Rbm24 through three distant enhancers, which are located in open chromatin regions and are bound by POU4F3. Lastly, Wang and colleagues discovered that ectopic expression of Rbm24 was unable to prevent the degeneration of POU4F3 null hair cells.

The findings in this manuscript hold great significance as they provide additional insights into the transcriptional cascades crucial for hair cell development. The discovery of enhancers capable of driving transgene expression specifically in hair cells holds promising therapeutic implications. The figures presented in the study are of excellent quality, the employed techniques are state-of-the-art, the data are accurately represented without exaggeration, and the study demonstrates a high level of rigor.

Reviewer #2 (Public Review):

Previous studies have shown that two hair cell transcription factors, Pou4f3 and Gfi1, are both necessary for the survival of cochlear hair cells, and that Gfi1 is regulated by Pou4f3. The authors have previously also shown that mosaic inactivation of the RNA-binding protein RBM24 leads to outer hair cell death.

In the present study, the authors show that hair cells die in Pou4f3 and Gfi1 mutant mice. They show that Gfi1 is regulated by Pou4f3. Both these observations have been published before. They then show that RBM24 is absent in Pou4f3 knockouts, but not Gfi1 knockouts. They ectopically activate RMB24 in the hair cells of Pou4f3 knockouts, but this does not rescue the hair cell death. Finally, the authors validate three RMB24 enhancers that are active in young hair cells and which have been previously shown to bind Pou4f3.

The experiments are well-executed and the data are clear. The results support the conclusions of the paper.

Much of the work in the paper has been reported before. The result that hair cell transcription factors operate in a network, with some transcription factors activating only a subset of hair cell genes, is an expected result. Since RMB24 is only one of many genes regulated directly by Pou4f3, it is not surprising that it cannot rescue the Pou4f3 knockout hair cell degeneration.

The identification of new hair cell enhancers may be of use to investigators wishing to express genes in hair cells.

In sum, this work, although carefully performed, does not shed significant new light on our understanding of hair cell development or survival.